

Problem Set 7

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1. Replicate both parts of Figure 1 for Bartel

```
# Import modules
library(tidyverse) # Load the tidyverse package for data wrangling and analysis
```

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.2.1      v readr      2.2.0
```

```
v forcats    1.0.1      v stringr    1.6.0
```

```
v ggplot2    4.0.3      v tibble     3.3.1
```

```
v lubridate  1.9.5      v tidyr      1.3.2
```

```
v purrr      1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```

library(haven) # Load haven so read_dta() can be used to import Stata data files
library(ggplot2) # Load ggplot2 for creating graphs and plots

# Read the ANES data
anes_2024 <- read_dta("~/Desktop/Current Classes/R/Problem Set 7/Data/anes_2024.dta") # Import

# Read the cumulative ANES data
anes_cum <- read_dta("~/Desktop/Current Classes/R/Problem Set 7/Data/anes_cum.dta") # Import

# Recode VCF0305 into four party identification categories

figure1 <- anes_cum |> # Start with the cumulative ANES dataset and create a new object called figure1
# Filter to have same years
filter(VCF0004 >= 1952 & VCF0004 <= 1996) |> # Keep only observations from 1952 through 1996
mutate( # Create a new variable called party_id
  party_id = case_when( # Assign a party identification category based on the value of VCF0305
    VCF0305 == 1 ~ "Pure Independents",
    VCF0305 == 2 ~ "Independent Leaners",
    VCF0305 == 3 ~ "Weak Identifiers",
    VCF0305 == 4 ~ "Strong Identifiers",
    # Assign missing values to observations that do not match any of the categories
    TRUE ~ NA_character_
  )
) |>
# Remove observations where party_id is missing
filter(!is.na(party_id))

# Calculate proportions

# Start with the figure1 dataset and overwrite it with the summarized data
figure1 <- figure1 |>
# Group the data by election year and party identification category
group_by(VCF0004, party_id) |>
# Count the number of respondents in each year and party category
summarize(n = n(), .groups = "drop") |>
# Group the data again by election year
group_by(VCF0004) |>
# Calculate each party category's proportion of respondents within each year
mutate(proportion = n / sum(n)) |>
# Remove the grouping structure from the data
ungroup()

```

```

# Start with figure1 and create the panel variable
figure1 <- figure1 |>
  mutate(
    panel = case_when( # Assign each party category to either the Identifiers or Independents
      party_id %in% c("Strong Identifiers", "Weak Identifiers") ~ "Identifiers",
      party_id %in% c("Independent Leaners", "Pure Independents") ~ "Independents"
    )
  )

# PLOTTING
ggplot(figure1, aes(
  # Put election year on the x-axis
  x = VCF0004,
  # Put the proportion of respondents on the y-axis
  y = proportion,
  # Group observations by party identification category
  group = party_id,
  # Use different line types for each party identification category
  linetype = party_id,
  # Use different point shapes for each party identification category
  shape = party_id
)) +
  geom_line() + # Draw lines connecting the observations across election years

  # Points only at labeled years
  geom_point( # Add points to the graph
    data = filter( # Use a filtered version of the data for the points
      figure1, # Start with the figure1 dataset
      VCF0004 %% 4 == 0 # Only show points for years divisible by 4
    ),
    size = 2 # Set the size of the points
  ) +

  # Create separate panels for Identifiers and Independents
  facet_grid(panel ~ ., scales = "fixed") +

  # Customize the y-axis
  scale_y_continuous(
    limits = c(0, 0.5), # Set the y-axis limits from 0 to 0.5
    breaks = seq(0.1, 0.5, 0.1) # Add y-axis tick marks every 0.1
  )

```

```

) +

# Customize the x-axis
scale_x_continuous(
  breaks = c(1956, 1964, 1972, 1980, 1988, 1996) #Specify which years should be labeled on
) +

# Manually set the line type for each party identification category
scale_linetype_manual(
  values = c( # Define the line type for each category
    "Strong Identifiers" = "solid",
    "Weak Identifiers" = "dashed",
    "Independent Leaners" = "dashed",
    "Pure Independents" = "solid"
  )
) +

# Manually set the point shape for each party identification category
scale_shape_manual(
  values = c( # Define the point shape for each category
    "Strong Identifiers" = 16,
    "Weak Identifiers" = 1,
    "Independent Leaners" = 1,
    "Pure Independents" = 16
  )
) +

labs( # Add labels to the graph
  # Remove the x-axis title
  x = NULL,
  # Remove the y-axis title
  y = NULL,
  # Graph title
  title = "Proportions of National Election Study Sample"
) +

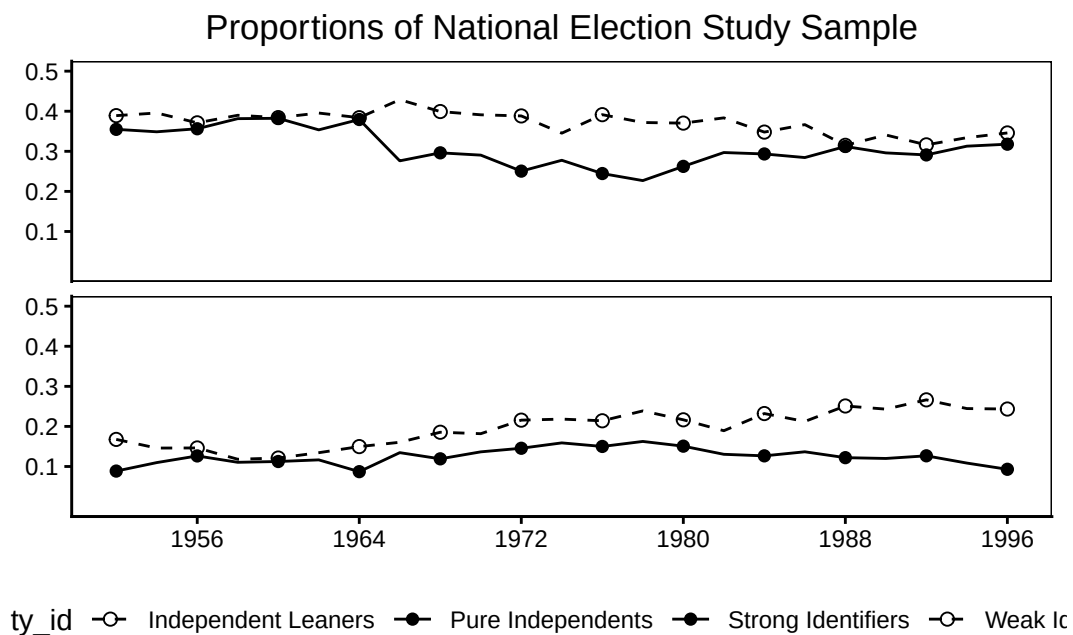
# Apply the classic ggplot theme
theme_classic() +
# Customize the appearance of the graph
theme(
  legend.position = "bottom",
  # Remove the background behind the facet labels

```

```

strip.background = element_blank(),
# Remove the facet labels
strip.text = element_blank(),
# Add a border around each panel
panel.border = element_rect(
  # Make the border black
  color = "black",
  # Do not fill the panel with a color
  fill = NA
),
# Customize the graph title
plot.title = element_text(
  # Center the graph title
  hjust = 0.5
)
)
)

```



3. Extend the analysis to 2024, using the 2024 ANES file. (This will require using the 2024 file in conjunction with the cumulative file.)

```
# Create the same four party identification categories
# for the 2024 ANES data

# Start with the 2024 ANES data
anes_2024 <- anes_2024 |>
  # Add new variables to the dataset
  mutate(
    # Create the election year variable and set it to 2024
    VCF0004 = 2024,
    # Create the party identification variable
    party_id = case_when(

      # Pure Independents
      V241221 == 3 & V241223 == 2 ~ "Pure Independents",

      # Independent Leaners
      V241221 == 3 & V241223 %in% c(1, 3) ~ "Independent Leaners",

      # Strong Identifiers
      V241221 %in% c(1, 2) & V241222 == 1 ~ "Strong Identifiers",

      # Weak Identifiers
      V241221 %in% c(1, 2) & V241222 == 2 ~ "Weak Identifiers",

      TRUE ~ NA_character_ # Assign missing values to respondents who do not fit any category
    )
  )

# Create the same party_id variable for the cumulative data

# Start with the cumulative ANES dataset
anes_cum_extended <- anes_cum |>
  # Create a new party_id variable
  mutate(
    # Assign party identification categories based on VCF0305
    party_id = case_when(
      VCF0305 == 1 ~ "Pure Independents",
      VCF0305 == 2 ~ "Independent Leaners",
      VCF0305 == 3 ~ "Weak Identifiers",
```

```

    VCF0305 == 4 ~ "Strong Identifiers",
    TRUE ~ NA_character_ # Assign missing values to observations that do not match a category
  )
)

# Combine cumulative ANES and 2024 ANES

# Combine the two datasets by stacking their rows
anes_all <- bind_rows(
  # Add the cumulative ANES observations
  anes_cum_extended,
  # Add the 2024 ANES observations
  anes_2024
)

# Calculate proportions

# Start with the combined ANES dataset
figure1_extended <- anes_all |>
  # Keep observations from 1952 through 2024
  filter(VCF0004 >= 1952 & VCF0004 <= 2024) |>
  # Remove observations with missing party identification
  filter(!is.na(party_id)) |>
  # Group respondents by election year and party identification
  group_by(VCF0004, party_id) |>
  # Summarize the number of respondents in each group
  summarize(
    # Count the number of respondents in each group
    n = n(),
    # Remove grouping after summarizing
    .groups = "drop"
  ) |>
  # Group the summarized data by election year
  group_by(VCF0004) |>
  # Create a new proportion variable
  mutate(
    # Calculate each party category's share of respondents for each year
    proportion = n / sum(n)
  ) |>
  # Remove the grouping structure
  ungroup() |>
  # Create a panel variable

```

```

mutate(
  panel = case_when( # Assign party categories to the appropriate panel
    # Check whether party_id belongs to either identifier category
    party_id %in% c(
      "Strong Identifiers",
      "Weak Identifiers"
    ) ~ "Identifiers",

    party_id %in% c( # Check whether party_id belongs to either independent category
      "Independent Leaners",
      "Pure Independents"
    ) ~ "Independents"
  )
)

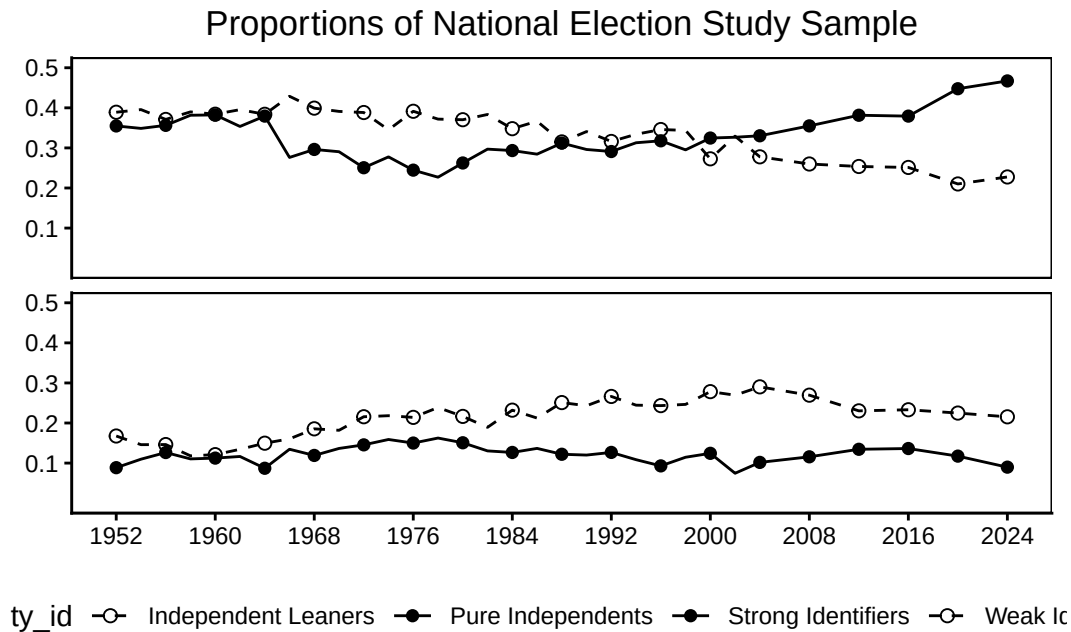
# Create a ggplot using the extended dataset
ggplot(
  figure1_extended,
  # Define the variables used in the graph
  aes(
    x = VCF0004,
    y = proportion,
    group = party_id, # Group observations by party identification category
    linetype = party_id, # Use different line types for each party identification category
    shape = party_id # Use different point shapes for each party identification category
  )
) +
  # Draw lines connecting the observations across years
  geom_line() +
  # Add points to the graph
  geom_point(
    # Only display points for years divisible by 4
    data = filter(figure1_extended, VCF0004 %% 4 == 0),
    # Set the point size
    size = 2
  ) +
  # Create separate panels for Identifiers and Independents
  facet_grid(panel ~ ., scales = "fixed") +
  # Customize the y-axis
  scale_y_continuous(
    limits = c(0, 0.5),
    breaks = seq(0.1, 0.5, 0.1)
  )

```

```

) +
# Customize the x-axis
scale_x_continuous(
  # Label the x-axis every 8 years from 1952 through 2024
  breaks = seq(1952, 2024, 8)
) +
scale_linetype_manual( # Manually specify line types
  values = c(
    "Strong Identifiers" = "solid",
    "Weak Identifiers" = "dashed",
    "Independent Leaners" = "dashed",
    "Pure Independents" = "solid"
  )
) +
scale_shape_manual(
  # Assign a point shape to each party identification category
  values = c(
    "Strong Identifiers" = 16,
    "Weak Identifiers" = 1,
    "Independent Leaners" = 1,
    "Pure Independents" = 16
  )
) +
# Add labels to the graph
labs(
  x = NULL,
  y = NULL,
  title = "Proportions of National Election Study Sample"
) +
# Apply the classic ggplot theme
theme_classic() +
theme(
  legend.position = "bottom", # Place the legend below the graph
  strip.background = element_blank(), # Remove the facet background
  strip.text = element_blank(), # Remove the facet labels
  panel.border = element_rect( # Add a border around each panel
    color = "black", # Set the border color to black
    fill = NA # Keep the inside of the panel unfilled
  ),
  plot.title = element_text(hjust = 0.5) # Center the graph title
)

```



4. Create figure that include midterm election years as well

```
# Create figure including presidential and midterm election years
# Use the cumulative ANES data

# Start with the cumulative ANES dataset
figure1_midterms <- anes_cum |>
  # Keep observations from 1952 through 2020
  filter(VCF0004 >= 1952 & VCF0004 <= 2020) |>
  # Create the party identification variable
  mutate(
    # Recode VCF0305 into four party identification categories
    party_id = case_when(
      VCF0305 == 1 ~ "Pure Independents",
      VCF0305 == 2 ~ "Independent Leaners",
      VCF0305 == 3 ~ "Weak Identifiers",
      VCF0305 == 4 ~ "Strong Identifiers",
      TRUE ~ NA_character_ # Assign missing values to observations that do not match a category
    )
  ) |>
```

```

# Remove observations with missing party identification
filter(!is.na(party_id)) |>
# Group observations by election year and party identification
group_by(VCF0004, party_id) |>
# Count the number of respondents in each group
summarize(n = n(), .groups = "drop") |>
# Group the data by election year
group_by(VCF0004) |>
# Calculate the proportion of respondents in each party category
mutate(proportion = n / sum(n)) |>
# Remove the grouping structure
ungroup() |>
# Create the panel variable
mutate(
  panel = case_when(
    party_id %in% c("Strong Identifiers", "Weak Identifiers") ~ "Identifiers",
    party_id %in% c("Independent Leaners", "Pure Independents") ~ "Independents"
  )
)

# Plot
ggplot(
  # Use the figure1_midterms dataset
  figure1_midterms,
  aes(
    x = VCF0004, # Put election year on the x-axis
    y = proportion, # Put the proportion on the y-axis
    group = party_id, # Group observations by party identification category
    linetype = party_id, # Use different line types for each party identification category
    shape = party_id # Use different point shapes for each party identification category
  )
) +

# Lines connect all available ANES years
geom_line() +

# Points show all available ANES years, including midterms
geom_point(size = 2) + # Add points

# Create two panels
facet_grid( # Divide the graph into two panels
  panel ~ ., # Place the panels vertically based on the panel variable

```

```

    scales = "fixed" # Use the same scale for both panels
  ) +

  # Y-axis
  scale_y_continuous( # Customize the y-axis
    limits = c(0, 0.5), # Set the y-axis limits from 0 to 0.5
    breaks = seq(0.1, 0.5, 0.1) # Add tick marks every 0.1
  ) +

  # X-axis
  scale_x_continuous( # Customize the x-axis
    breaks = seq(1952, 2020, 4) # Label the x-axis every 4 years from 1952 through 2020
  ) +

  # Line types
  # Manually set the line type for each party identification category
  scale_linetype_manual(
    # Define the line type for each category
    values = c(
      "Strong Identifiers" = "solid",
      "Weak Identifiers" = "dashed",
      "Independent Leaners" = "dashed",
      "Pure Independents" = "solid"
    )
  ) +

  # Point shapes
  scale_shape_manual(
    values = c(
      "Strong Identifiers" = 16,
      "Weak Identifiers" = 1,
      "Independent Leaners" = 1,
      "Pure Independents" = 16
    )
  ) +

  # Labels
  labs(
    x = NULL, # Remove the x-axis title
    y = NULL, # Remove the y-axis title
    title = "Proportions of National Election Study Sample" # Add the graph title
  ) +

```

```

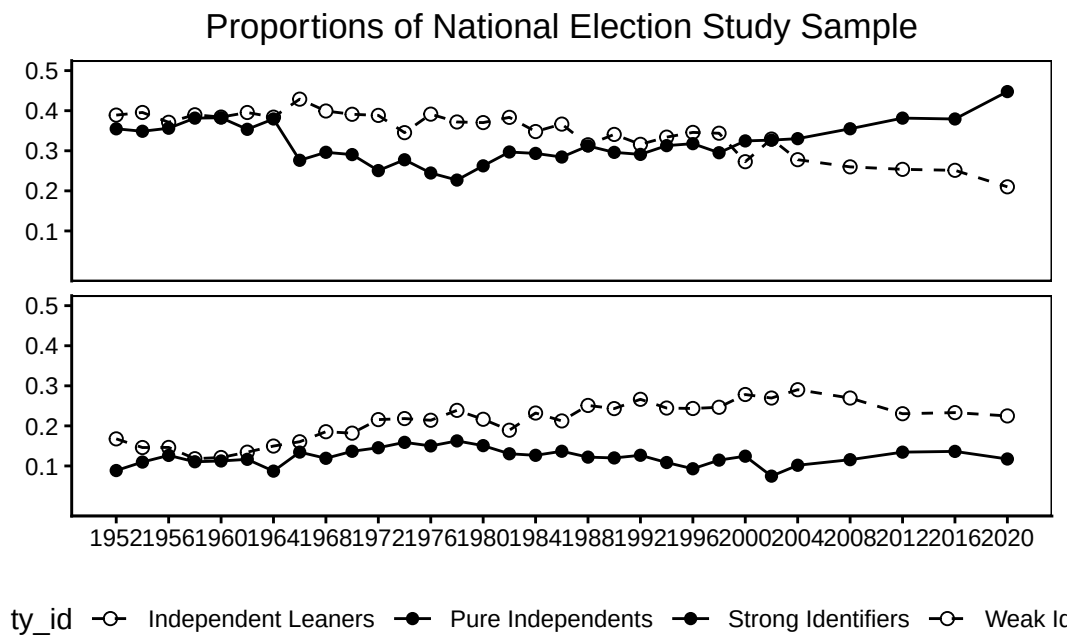
# Theme
# Apply the classic ggplot theme
theme_classic() +

theme(
  legend.position = "bottom", # Place the legend underneath the graph
  strip.background = element_blank(), # Remove the background behind facet labels
  strip.text = element_blank(), # Remove the facet labels

  # Border around each panel
  panel.border = element_rect(
    color = "black", # Make the panel border black
    fill = NA # Do not fill the panel with a color
  ),

  # Center title
  plot.title = element_text( # Customize the graph title
    hjust = 0.5 # Center the title horizontally
  )
)
)

```



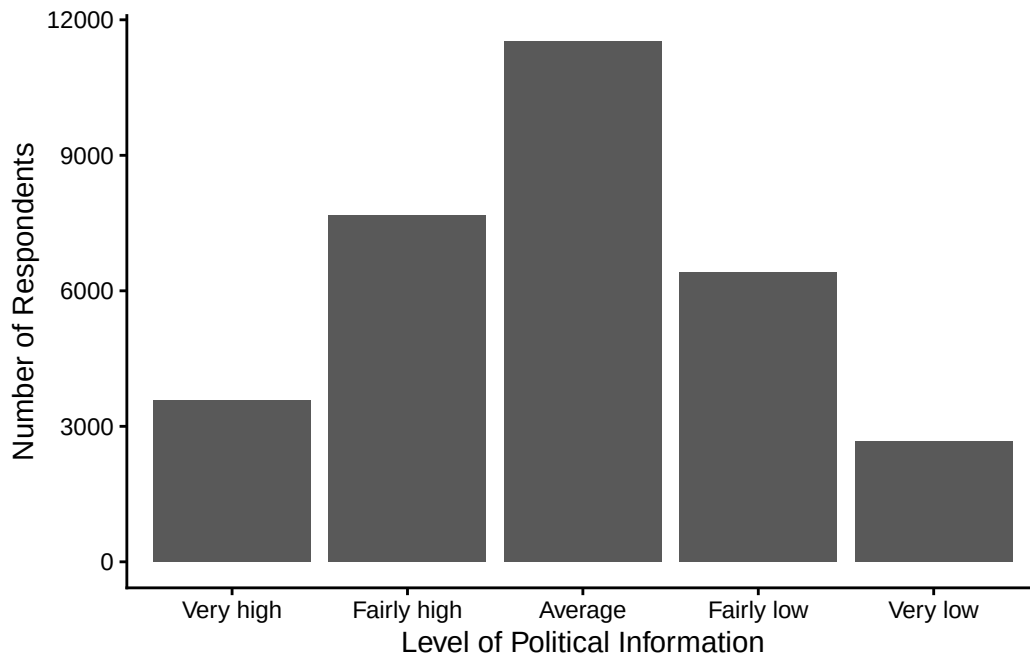
5. Original Work

```
# Original Work

# Start with the cumulative ANES dataset
political_info <- anes_cum |>
  # Create a new variable called political_info
  mutate(
    # Recode
    political_info = case_when(
      VCF0050b == 1 ~ "Very high",
      VCF0050b == 2 ~ "Fairly high",
      VCF0050b == 3 ~ "Average",
      VCF0050b == 4 ~ "Fairly low",
      VCF0050b == 5 ~ "Very low",
      TRUE ~ NA_character_ # Assign missing values to observations that do not match a category
    )
  ) |>
# Remove observations with missing political information
filter(!is.na(political_info)) |>
# Modify the political_info variable
mutate(
  # Convert political_info from text to a factor
  political_info = factor(
    # Specify the variable being converted to a factor
    political_info,
    # Set the order of the factor levels
    levels = c(
      "Very high",
      "Fairly high",
      "Average",
      "Fairly low",
      "Very low"
    )
  )
)
# Plot

# Create a graph with political information categories on the x-axis
ggplot(political_info, aes(x = political_info)) +
  # Create a bar chart showing the number of respondents in each category
  geom_bar() +
```

```
# labels
labs(
  x = "Level of Political Information",
  y = "Number of Respondents"
) +
theme_classic() # Apply the classic ggplot theme
```



```
# VCF0504 - the republicans party liberal to conservative scale

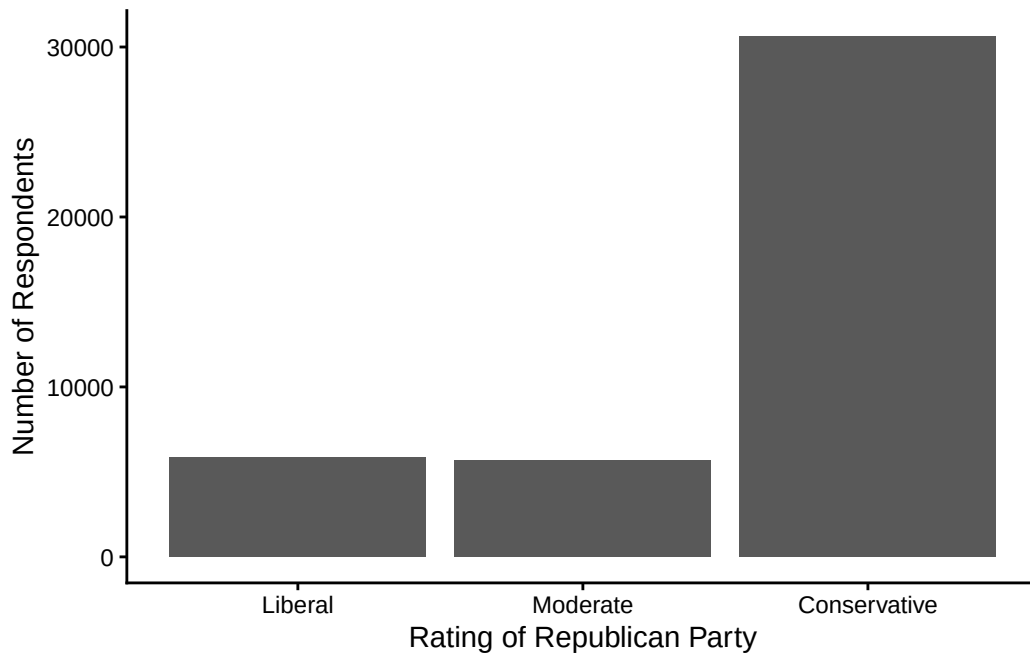
# Start with the cumulative ANES dataset
republican_ideology <- anes_cum |>
# Create a new variable called republican_ideology
mutate(
  # Recode
  republican_ideology = case_when(
    VCF0504 %in% c(1, 2, 3) ~ "Liberal",
    VCF0504 == 4 ~ "Moderate",
    VCF0504 %in% c(5, 6, 7) ~ "Conservative",
    TRUE ~ NA_character_ # Assign missing values to observations that do not match a category
  )
) |>
# Remove observations with missing ideology categories
```

```

filter(!is.na(republican_ideology)) |>
# Modify the republican_ideology variable
mutate(
  # Convert republican_ideology into a factor
  republican_ideology = factor(
    # Specify the variable being converted
    republican_ideology,
    levels = c(
      "Liberal",
      "Moderate",
      "Conservative"
    )
  )
)

# Create a graph with Republican ideology categories on the x-axis
ggplot(republican_ideology, aes(x = republican_ideology)) +
  # Create a bar chart showing the number of respondents in each category
  geom_bar() +
  # labels
  labs(
    x = "Rating of Republican Party",
    y = "Number of Respondents"
  ) +
  # Apply classic theme
  theme_classic()

```



Follow-up Questions

Where did you encounter problems with this replication? Was there anything that was particularly tricky?

The most difficult part was making sure the party identification categories were coded consistently with the original figure, especially when working with the different variables in the cumulative ANES file and the 2024 ANES file. It was also tricky to combine the 2024 data with the cumulative data because the variables used to measure party identification are different between the two files.

###How would you interpret (explain) what Bartels found in figures 1 & 2?

Bartels' Figure 1 shows a clear decline in strong party identification and an increase in independent identification between 1952 and 1996. The important point is that this does not mean Americans simply stopped having partisan preferences; much of the movement was toward weaker forms of party identification, especially independent leaners. In the top panel, the share of strong party identifiers was around 35% in the 1950s and early 1960s, but fell sharply after 1964, reaching roughly 23–25% in the 1970s. It recovered somewhat by the 1990s but remained below its earlier level. In the bottom panel, independent leaners rose from roughly 15% in the 1950s to about 25–27% by the late 1980s and early 1990s.

###Has anything changed since 1996 that would alter the substantive takeaway points in their findings?

After 1996, the proportion of “Strong” Identifiers generally increased, while the proportion of “Weak” Identifiers decreased. At the same time, both “Pure” Independents and Independent “Leaners” generally declined. This suggests a reversal of the trend Bartels observed in the earlier period: after 1996, Americans appear to have become more strongly attached to political parties, while fewer identified as independents.

###Does anything change about the analysis in a substantive way when you include midterm election years?

Midterm years add extra data points and short-term noise, but they do not alter the main story. Midterm years just reflect lower voter turnout, which creates temporary bumps because midterm voters are slightly more partisan. Whether you look at midterms or not, the main idea is identical, with a big drop in party loyalty in the 1960s-1970s, followed by a huge comeback in party loyalty after 1996.